

IPOH, Malaysia

WMO: 486250

Lat: 4.5663N

Lon: 101.0930E

Elev: 40

StdP: 100.85

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 11	(b) 22.2	(c) 22.8	(d) 19.0	(e) 13.9	(f) 27.1	(g) 19.9	(h) 14.7	(i) 27.3	(j) 5.5	(k) 27.6	(l) 4.9	(m) 27.8	(n) 1.8	(o) 40	(p) 0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 8.5	(c) 35.0	(d) 26.1	(e) 34.2	(f) 26.3	(g) 34.0	(h) 26.3	(i) 28.0	(j) 32.7	(k) 27.6	(l) 32.3	(m) 27.3	(n) 31.9	(o) 2.6	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.7	31.2	26.2	21.7	30.3	26.1	21.6	30.2	90.0	32.3	88.1	32.9	86.7	32.0	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.2	(b) 4.4	(c) 3.9	(e) 21.5	(f) 36.0	(g) 0.8	(h) 0.8	(i) 21.0	(j) 36.5	(k) 20.5	(l) 37.0	(m) 20.1	(n) 37.4	(o) 19.5	(p) 38.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			19.6	28.8	1.0	0.5	18.9	29.2	18.3	29.4	17.7	29.7	17.0	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	27.9	28.6	28.9	28.8	28.8	28.8	28.6	28.4	28.2	28.0	27.9	27.7
	DBStd	1.00	1.07	0.94	1.06	0.93	0.78	0.90	0.99	0.91	0.91	0.76	0.79	0.95
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6708	556	520	584	564	584	563	576	571	545	559	537	550
	CDD18.3	3667	297	287	326	314	326	313	317	313	295	301	287	292
	CDH23.3	35567	2634	2849	3266	3111	3432	3410	3314	3170	2819	2699	2385	2477
	CDH26.7	14309	1024	1243	1414	1279	1402	1469	1383	1306	1087	999	832	871
(14) Wind	WSAvg	1.7	1.7	1.7	1.7	1.6	1.6	1.7	1.7	1.7	1.7	1.7	1.5	1.6
(15) Precipitation	PrecAvg	2387	150	131	166	250	213	146	162	156	208	297	274	236
	PrecMax	3068	413	294	339	602	469	390	346	313	389	581	497	438
	PrecMin	1913	15	20	20	96	75	29	39	50	53	92	70	81
	PrecStd	268	87	72	71	94	99	80	78	79	83	114	90	100
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	35.8	36.0	35.2	34.6	35.0	35.1	34.9	34.2	34.1	33.8	34.1
		MCWB	25.9	24.7	26.0	26.9	27.1	26.1	26.0	25.9	25.8	26.3	26.6	26.7
	2%	DB	33.2	34.2	34.9	34.1	34.0	34.1	34.0	33.9	33.2	33.1	33.0	33.1
		MCWB	25.8	25.7	26.1	27.2	27.1	26.3	26.2	25.8	25.8	26.2	26.5	26.1
	5%	DB	32.9	33.9	34.0	33.2	33.1	33.2	33.1	33.1	32.8	32.2	32.1	32.2
		MCWB	25.8	25.8	26.4	27.0	27.0	26.4	26.1	25.8	25.8	26.1	26.4	25.9
	10%	DB	31.9	33.0	33.0	32.8	32.8	32.9	32.2	32.1	31.9	31.8	31.2	31.2
		MCWB	25.6	25.7	26.3	26.9	26.9	26.3	26.0	25.7	25.8	26.0	26.4	26.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.3	27.7	27.9	28.4	28.5	27.9	28.0	27.7	27.2	27.9	27.9	27.6
		MCDB	32.3	32.6	33.4	33.3	33.0	32.6	32.8	31.9	31.5	32.3	32.1	32.4
	2%	WB	26.7	27.1	27.3	27.7	27.9	27.4	27.2	26.8	26.7	27.0	27.2	26.8
		MCDB	31.5	32.1	32.5	32.5	32.3	31.9	31.7	31.3	31.1	31.5	31.5	31.3
	5%	WB	26.4	26.6	26.9	27.3	27.5	27.0	26.7	26.4	26.4	26.6	26.7	26.4
		MCDB	31.0	31.5	32.0	32.0	31.8	31.4	31.1	30.9	30.7	30.8	30.8	30.7
	10%	WB	26.0	26.2	26.5	26.9	27.0	26.6	26.3	26.1	26.1	26.2	26.2	26.0
		MCDB	30.4	30.8	31.3	31.3	31.2	30.9	30.7	30.6	30.4	30.2	30.1	30.1
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	9.6	9.3	8.8	8.3	8.5	8.5	8.6	8.3	8.3	8.2	8.0
		MCDBR	9.8	10.6	10.2	9.3	8.8	9.1	9.2	9.3	9.1	8.9	8.9	9.2
		MCWBR	4.1	4.2	4.0	3.9	3.8	3.8	3.7	3.6	3.6	3.6	3.8	3.8
	5% WB	MCDBR	9.1	9.5	9.3	8.9	8.3	8.5	8.6	8.7	8.3	8.4	8.5	8.4
		MCWBR	4.0	4.1	4.0	3.9	3.8	3.8	3.8	3.7	3.6	3.6	3.7	3.7
(40) Clear-Sky Solar Irradiance	taub		0.457	0.501	0.506	0.511	0.499	0.526	0.516	0.529	0.529	0.540	0.467	0.444
	taud		2.299	2.156	2.161	2.154	2.199	2.124	2.148	2.100	2.103	2.069	2.290	2.358
	Ebn at Noon		866	836	830	811	800	766	779	785	799	792	850	872
	Edh at Noon		135	159	158	156	145	154	152	163	165	171	135	125
(44) All-Sky Solar Radiation	RadAvg		4.66	5.44	5.54	5.53	5.17	5.12	5.02	5.10	4.93	4.74	4.49	4.27
	RadStd		0.36	0.37	0.42	0.17	0.18	0.22	0.22	0.18	0.27	0.35	0.20	0.33

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.39	+0.22	N/A	+0.52	+0.44	+0.37	N/A	N/A	+140	+140

Nomenclature: See separate page